

Advancing Knowledge Together: Integrating Large Language Model-based Conversational AI in Small Group Collaborative Learning

Zhenyao Cai*

zhenyaoc@uci.edu

University of California, Irvine
Irvine, California, USA

Nia Nixon

dowelln@uci.edu

University of California, Irvine
Irvine, California, USA

Seehee Park*

seeheep@uci.edu

University of California, Irvine
Irvine, California, USA

Shayan Doroudi

doroudis@uci.edu

University of California, Irvine
Irvine, California, USA

ABSTRACT

In today's educational landscape, students learn collaboratively, where students benefit from both peer interactions and facilitator guidance. Prior research in Human-Computer Interaction (HCI) and Computer-Supported Collaborative Learning (CSCL) has explored chatbots and AI techniques to aid such collaboration. However, these methods often depend on predefined dialogues (which limits adaptability), are not based on collaborative learning theories, and do not fully recognize the learning context. In this paper, we introduce an Large Language Model (LLM)-powered conversational AI, designed to enhance small group learning through its advanced language understanding and generation capabilities. We detail the iterative design process, final design, and implementation. Our preliminary evaluation indicates that the bot performs as designed but points to considerations in the timing of interventions and bot's role in discussions. The evaluation also reveals that learners perceive the bot's tone and behavior as important for engagement. We discuss design implications for chatbot integration in collaborative learning and future research directions.

CCS CONCEPTS

• **Applied computing** → **Collaborative learning; Interactive learning environments**; • **Human-centered computing** → Collaborative and social computing systems and tools; Empirical studies.

KEYWORDS

Collaborative Learning, Human-AI Collaboration, AI facilitator

*Both authors contributed equally to this research.

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650868>

ACM Reference Format:

Zhenyao Cai, Seehee Park, Nia Nixon, and Shayan Doroudi. 2024. Advancing Knowledge Together: Integrating Large Language Model-based Conversational AI in Small Group Collaborative Learning. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3613905.3650868>

1 INTRODUCTION

A substantial body of research has been conducted to explore and validate the advantages of collaborative learning in groups [16, 17]. Prior research suggests that collaborating in groups offers multiple benefits [26, 34], not only knowledge gain and improved academic performance, but also enriched social interactions, and psychological well-being.

Past studies have shown that group facilitators play an important role in orchestrating effective knowledge sharing and construction within the student groups [12]. It is necessary to have a skilled facilitator to guide the conversation direction, pose questions to stimulate new ideas, foster a culture of diverse opinions, set the tone of the discussion, and steer the conversation back on track when digressions occur. However, guiding the direction of a conversation as a student facilitator can be challenging due to several factors, including a lack of facilitation experience and skills, power dynamics and authority within the group, and the increased cognitive load for multitasking.

In light of this, recent research has explored various conversational-based techniques and tools to support and facilitate group work [14, 16, 25] and collaborative learning [6, 15, 31, 35, 38]. However, prior works used predefined dialogues that have limited adaptability, not fully considering the potential of context-sensitive dialog systems that can learn and personalize interactions for enhanced user engagement across different groups and learning scenarios. The recent advancement of LLM can potentially remedy this situation given the ability of LLMs to generate highly contextual responses and naturally adapt to the flow of a conversation. This might enable the creation of chatbots that can offer personalized and dynamic interactions with minimal coding requirements [9, 42].

In this study, we explore the design and development of a chatbot facilitator powered by LLM to facilitate collaborative learning during group discussions. We also present preliminary results from

an evaluation conducted with a small group of three undergraduate students. The primary goal of this study is to introduce an adaptable collaborative learning platform incorporating a LLM-powered bot. This platform aims to be scalable for future experiments testing hypotheses about the role of conversational AI in collaborative learning environments.

2 RELATED WORK

AI chatbots have been used to facilitate group work in Human-Computer Interaction (HCI) and Computer-Supported-Cooperative-Work (CSCW). Prior work in HCI on chatbot-facilitated collaboration has explored how AI can enhance group dynamics and social interactions. For example, Shin et al. [36] developed IntroBot to enhance familiarity between members, Benke et al. [8] designed chatbot for group emotion management, and Kim et al. [25] investigated how chatbots could facilitate equal participation.

Despite these advances in HCI and CSCW, the exploration of AI chatbots within collaborative learning settings remains underexplored. Existing research on chatbots in educational settings predominantly focuses on one-to-one learning, where a bot interacts with a single student. AutoTutor is an example of such a system, designed for a bot tutor to teach and prompt students to learn and think more deeply about a learning topic [18]. Studies by Huang et al. and Jin et al. [21, 23] highlighted the potential of chatbots in improving learning outcomes by providing individualized scaffolding and feedback. However, less research has been conducted on how a bot could tutor and facilitate in a group setting. Survey papers by Tan et al. [37] and Nixon et al. [33] suggest that AI could positively impact group learning by encouraging collaborative learning through engaging dialogues that enhance problem-solving, articulation, reflection, and idea sharing. A more recent study by Ha [31] investigated the role of bot facilitators for collaborative learning in small groups and their effect on group interactions and learning outcomes. The major limitation of prior research on these chatbots is their lack of context-specific adaptability due to rule-based chatbot design. With advances in LLMs' language understanding and generation capabilities, our study is the first to investigate the potential of LLM in facilitating small group collaborative learning with context-specific human-like facilitation behaviors, adaptability to various learning tasks and scenarios, and minimal coding requirements for testing various empirical collaborative learning frameworks.

3 EDUCHAT PLATFORM

3.1 Design process

3.1.1 Design Goals. This chatbot is designed to facilitate group discussion and collaborative learning with three primary objectives. Firstly, drawing from previous AI research in group facilitation, the chatbot aims to promote inclusivity and balanced participation [14, 25]. The bot will alert the group when certain individuals dominate the conversation and gently remind quieter members to contribute. Secondly, unlike most previous work, our chatbot focuses on guiding the group through stages of discussion, ensuring the completion of collaborative tasks. We aim to facilitate transitions through stages following Tuckman's stages of group

development theory [39], moving from orientation, to open discussion, and finally to conclusion. Thirdly, a unique aspect of our approach to collaborative learning involves facilitating knowledge construction. This aims to promote deeper conversations and more constructive dialogue, building on prior frameworks. Specifically, we utilize the Interactive, Constructive, Active, and Passive (ICAP) framework [11] and the Interaction Analysis Model (IAM) [19, 32], which are empirical-based models for understanding knowledge construction phases and cognitive engagement in conversations.

3.1.2 Wizard-of-Oz Method. To understand the desired behaviors of the chatbot and the optimal timing for chiming in, we used the Wizard-of-Oz method [13]. In this setup, two researchers simulated the chatbot, adhering to the three proposed design goals, in discussions with three undergraduate computer science students at a research university in the United States. The process began with a pre-test survey asking for domain knowledge, confidence in discussing AI ethics, and attitudes towards collaboration and the use of chatbots in group discussions. The participants then engaged in two discussion tasks, first tackling a well-structured question about solving a programming problem and then an ill-structured question about the knowledge required for developing ethical and effective AI applications (see Appendix). After this, participants completed a post-test survey similar to the pre-test survey, followed by a semi-structured focus group discussion about their thoughts on chatbot design.

We coded each of the student's utterances using the ICAP framework to understand student engagement [11]. We further analyzed their responses to the open-ended question and the discussions during the focus group interviews to gain insight into their views about the "bot". The results highlighted three key findings: 1) All students preferred the chatbot's stage transition behaviors, including initiating and wrapping up discussions. 2) Certain knowledge construction facilitation behaviors, such as asking follow-up questions, posing doubt or disagreement, and building connections between students' ideas, might lead to more in-depth discussions. 3) Directly addressing quieter individuals made the participant discouraged to participate, as aligned with previous studies [14]. Instead, providing a general and friendly reminder to the whole group about participation might be more effective.

3.1.3 Iterative Design. To tailor our design to the needs of our target users – high school and undergraduate students – we adopted a participatory design approach. This involved five undergraduate students, including three from the Wizard-of-Oz study, in the design and implementation process. Over two months, we held weekly meetings to discuss design goals and design decisions.

3.2 EduChat Finalized System Features

Instead of integrating the chatbot into an existing chat platform, such as WhatsApp, Telegram [25], and Slack [43], we decided to implement a full web-based platform. This decision was guided by two considerations. First, it has freedom for implementing new functionalities. Existing platforms often come with inherent limitations, including restrictions on customization, data access, and integration with external tools. Building a web-based platform allows for better adaptability, enabling teachers to create rooms with

customized discussion tasks and modified bots. Second, this approach aligns with our goal to use the platform as a prototype for future research in collaborative learning with AI. By building our own platform, we can experiment with and refine different chatbot design guided by collaborative learning theories and chatbot design frameworks.

The system has two components, the bot and the web platform, which we will describe in more detail below.

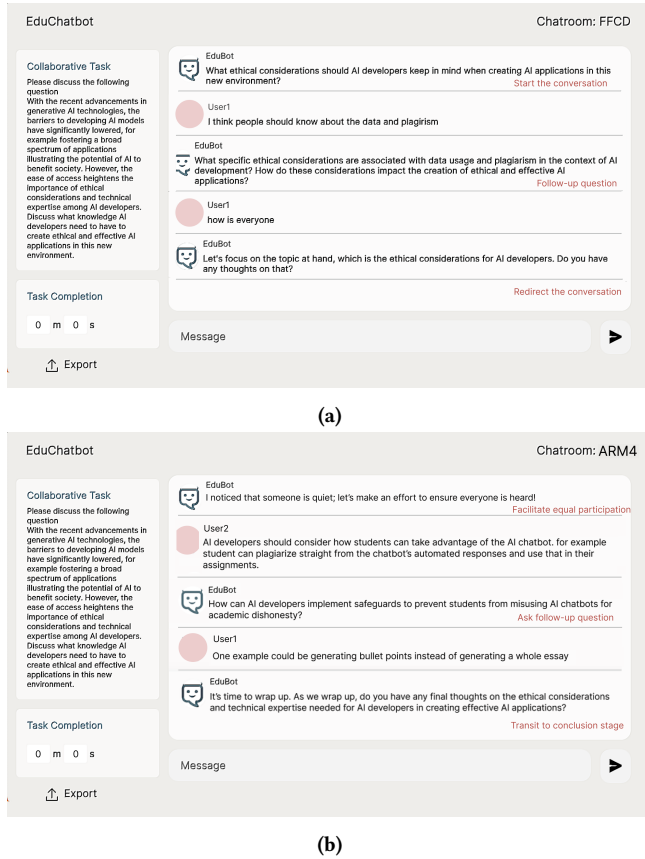


Figure 1: Chatroom Interface featuring the Bot: Bot behaviors include opening up the conversation, asking follow-up questions, directing the conversation when it goes off-topic, reminding the group about equal participation, and prompting the group to wrap up. Note that to clearly present bot behaviors, we have trimmed some messages in the original conversation.

3.2.1 Bot Features. Based on the theories and the participatory design process with undergraduate students, our bot is designed with three primary functionalities (Figure 1). The first is to facilitate the transition through phases of task completion, following Tuckman’s theory of stages in group discussion [39]. This includes initiating the discussion by posing a question, transitioning to sharing opinions, and finally moving towards conclusion by reminding participants of the time and encouraging them to wrap up their

discussions. Secondly, during the discussion phase, the bot is designed to ask follow-up questions, challenge students’ ideas, or make connections between the ideas of two or more students. This is intended to promote deeper conversations and more constructive dialogue, drawing upon insights from prior research in collaborative learning in small groups [11, 19, 32, 41]. Additionally, the bot is designed to reiterate the topic when the conversation strays off-topic. Finally, the bot is designed to remind the group about inclusivity and equal participation, by prompting them to give more opportunities to less-contributing participants.

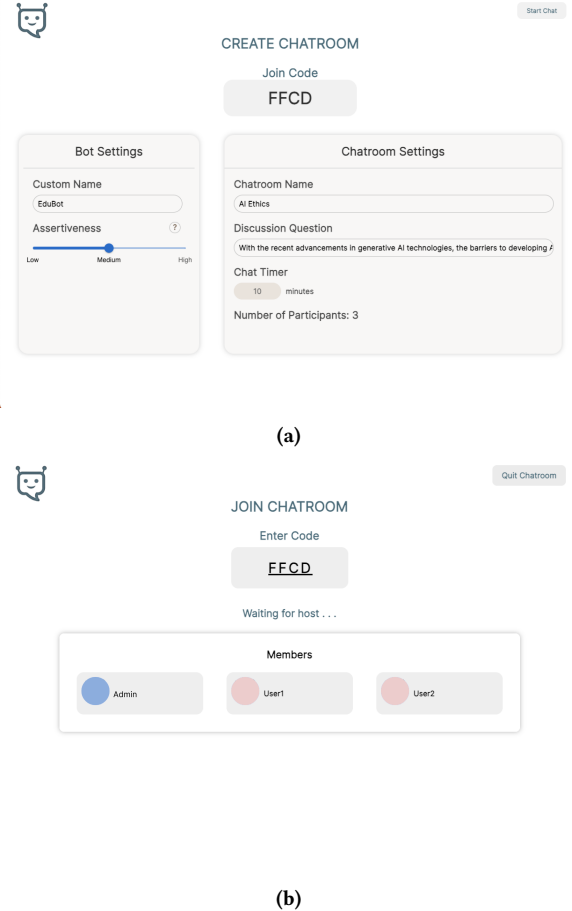


Figure 2: Platform Interface. (a) Page for teachers to create a room and customize the discussion task and the bot name and chime-in frequency; (b) Page for students to join the room by entering the unique room code.

3.2.2 Platform Features. The web platform featuring our chatbot has three main components, catering to both educators and students for an interactive experience (Figure 2). 1) The first component is the room creation page. This is for teachers or moderators to create a customized discussion room. They can set up a discussion room with a customized discussion topic, task, duration, and the bot’s name. This offers adaptability across various topics and provides teachers with more control and monitoring over the discussion.

2) The second feature is the room joining page for students. They can join their discussion room by entering a room code. This is designed for maintaining a secure discussion room that only students with the code can join. 3) The third component is the discussion chat room page, where joined students can see the discussion topic and task, send and read messages, and interact with the chatbot. Additionally, it includes an 'export conversation' function for teachers and researchers, allowing them to download the chat log for assessment and analysis.

3.3 Implementation

3.3.1 Bot Implementation. We implemented the bot in a layered manner (Figure 3). First, we employed a rule-based mechanism for stage transitions based on the time elapsed in the discussion. The bot greets students once they join and reminds them to transition to the conclusion when one-quarter of the discussion time remains. During the open discussion stage, the system tracks each message a user sends, maintaining a participation score for each participant and adding one point for every message they send. After each message, the system assesses whether participation is balanced. Our current mechanism is simple: when the highest-scoring participant reaches 3 points, it triggers an alert if any participant's participation falls below 10% of the total conversation. If this imbalance is not observed, the system proceeds to the chime-in layer. This layer sends the message history in the discussion room to the GPT-3.5 API [3] to determine whether it should intervene. The chime-in is designed to activate when the discussion is unproductive or when a student's idea merits deeper exploration. If the chime-in layer is triggered, the process moves to the behavior layer. Here, GPT determines the type of response needed to facilitate the conversation, guided by prompts we iterated upon during the design process (based on the framework and testing). We use few-shot prompting with examples designed together with undergraduate students during design iteration process. We want to note that our current prompt is not perfect, and further iteration is needed.

3.3.2 Web-app Implementation. In developing our chat application, we adopted a client-server architecture, with the server built on NodeJS [2] and Express, and the client developed using ReactJS [4]. We have also used Socket.io [5] for real-time discussion which facilitates bi-directional, event-driven communication. For creating customized and secure chatrooms, each chatroom is assigned a unique code, randomly generated when a user creates a new room. Alongside this, we have incorporated a timer feature for each room and task and topic customization for adaptability. In each room, we also implemented a unique instruction for calling the GPT-3.5 API [3] based on the creator's customization. This allows the chatbot to engage effectively with the conversation to provide contextually appropriate responses. Furthermore, we have used Firebase [1] for storing conversation histories. Each room's chat history is stored and organized separately by the room code. The creator of the chatroom can export the chat histories into CSV files for record-keeping and further analysis.

4 PRELIMINARY USER STUDY

We conducted a preliminary user study with a group of three undergraduate students to test our system and investigate the following questions:

- (1) Does the current design and implementation of the chatbot fulfill the three design goals (i.e. stage transition, equal participation, and knowledge construction)?
- (2) How do students perceive having such a chatbot in their discussion?

4.1 Participants and Procedure

The study was structured as follows: First, participants completed a pre-survey to assess their familiarity with generative AI technologies and their attitudes towards group collaboration and AI integration in discussions, and their thoughts on an ill-structured question about generative AI ethics, which was to be used for the discussion task (see Appendix). Following this, one researcher created a discussion room using the system with the same ill-structured discussion question and asked the three participants to join the chatroom. The participants then discussed the topic for 20 minutes with the bot's facilitation. After this, they filled out a post-survey identical to the pre-survey to gauge any shifts in their responses to the discussion question, their understanding of generative AI, and their perceptions of AI's role in discussions. Finally, we concluded the study with a focus group interview for 20 minutes. In this session, we asked participants about the perceived advantages and disadvantages of our system and soliciting suggestions for improvement. The entire session was audio-recorded and the study was conducted under the school's IRB guidelines.

4.2 Results

4.2.1 RQ1. Evaluating the Bot's Behaviors. Overall, **the bot performed as anticipated**. Stage transition of the bot: it initiated interactions by greeting participants and then reminded them to wrap up their discussions. In terms of promoting inclusivity, it prompted the team to engage less active participants when they remained silent for a long time. Regarding knowledge construction, it posed follow-up questions based on the students' responses. Nevertheless, during the discussion, **we did not observe the bot challenging the students' ideas or building connections between their ideas** as we had intended. This partially suggests that the model might have a bias in its responses and some instability.

However, there are several considerations to note: First, all participants mentioned that **the bot should find the proper time to chime in during open discussions**. As designed, we allowed the LLM to recognize the timing based on the designed prompt instruction and students' conversation, so it is not supposed to intervene when the discussion is deemed "productive". However, two participants preferred the chatbot to respond quicker, with one saying, "[I hope] the chatbot could respond more quickly. Sometimes it didn't respond to our conversation, and it took a minute to respond to things" (P1). On the other hand, one participant expressed a desire for the chatbot to intervene more slowly, allowing the students more autonomy in leading the discussion, saying, "I feel like giving a little bit more time for the [bot] to chime in. Because I think now the bot asks a [follow-up] question immediately after people. I feel like

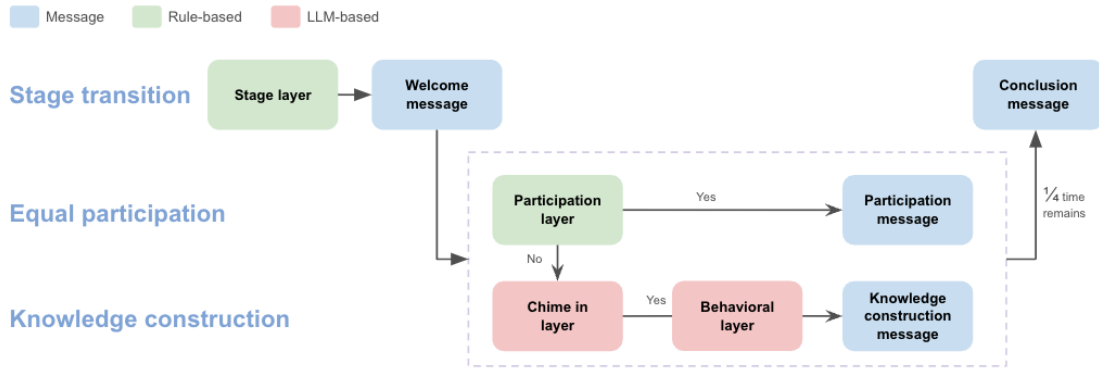


Figure 3: Bot Workflow

I'd rather stick to one question at a time and discuss and then move on" (P2). This suggests that careful consideration and a mechanism for determining when and how the bot should intervene in the discussion phase should be iterated in the future.

Second, two participants proposed that the **bot could act in various roles**, not only as a facilitator but also as a teacher or a helper. This role would involve assisting with answering questions or providing feedback. P3 mentioned, *"It could probably give a starting point to think about something. Sometimes, if you're clueless about a question, having some kind of answer helps you think more."* This idea was also raised by P2, who expressed a desire for the bot to provide feedback on their responses, saying, *"[I wish it could] assess our responses, because it was really just us answering the questions and not getting any feedback."*

4.2.2 RQ2. Participants' Perceptions of the Bot. Participants' **attitudes towards the bot were generally positive**, for two main reasons. First is the **bot's tone**. One participant mentioned that the professional tone of the bot reminded them to use more formal language instead of colloquial, saying, *"I also think this [reminded] me to be more formal, or more academic or use [professional] language"* (P3). Meanwhile, two other students mentioned that they liked the **bot's behavior**, which made them feel *"more inclusive and want to participate more"* (P1, P2) and *"want to delve deeper into the question"* (P2). P2 specifically noted perceiving the bot as very different from her classmates; with them, she *"asked a question and just got a brief response"*, whereas with the bot, it was more *"[back and forth], and having a deeper conversation."*

In the evaluation study, we also observed a shift in the participants' responses from the pre-survey to the post-survey. In the pre-survey, participants showed a wide range of ideas in answering open-ended questions. Their responses included concerns over *"creative theft"* associated with AI, the importance of considering AI's limitations in academic settings, and the need to be *"familiar with laws and regulations."* However, after students interacting with the chatbot, we noticed a convergence in their responses in the post-survey, predominantly focusing on *"federal law and regulations,"* a topic specifically raised by the chatbot. This change might be explained by the groupthink theory [22], which suggests that

decision-making in a group may discourage creativity. This might also imply that the chatbot could influence the participants' thought processes and perspectives, potentially shaping the direction of the discussion.

5 DISCUSSION

5.1 Design Implications for LLM-powered Chatbot for Collaborative Learning

In our study, we found that students' ideas were more diverse in the pre-survey, while all of them mentioned similar ideas in the post-survey after discussing together with the bot (and the idea was partially raised by the bot). Though our evaluation sample is small, this might imply about the students' trust and reliance on the AI's output for their learning, which could have implications for their learning process. Looking ahead, future systems could employ mechanisms and design elements that could **raise students' awareness and autonomy while using the bot** integrating their own values and perspectives into their responses.

Moreover, we found that students have different preferences regarding when and how the bot intervenes in the discussion. Considering the complexity and nuances of learning, future designs could consider **personalizing the bot in terms of behaviors, tones, and timing**, taking into account the subject, context, and the students' characteristics. With the LLM's capabilities for language understanding and generation, future designs could utilize this advantage to build a more adaptable chatbot that can be used in various contexts and integrated into real learning scenarios.

5.2 Implications for Future Research

First, building on prior research on learning dashboards that quantify and visualize individual and group performance and participation [29], we plan to integrate these with the LLM pipeline. By incorporating quantified measures of engagement and participation into the LLM pipeline, the LLM might be able to make more accurate decisions using a combination of rule-based and model-based approaches, thereby enhancing the precision of the decision-making process. Additionally, visualizing these metrics will make the bot's

decision-making process more transparent to users and inform participants about their performance and involvement.

Secondly, the platform serves as a valuable tool for testing various hypotheses related to students social dynamic and collaborative learning. In the study, our findings suggest that students might have perceived the bot's tone as impacting their enjoyment during the collaborative discussion task. In our future work, we plan to examine the potential effects of the bot's tone on discussion dynamics, learning behaviors, and results. Drawing inspiration from Ceha et al.'s work [10] on the impact of a humorous bot on learning, future work can explore these aspects in group learning settings. Furthermore, we also found that students have different preferences towards the bot's roles in their learning, such as being a teacher, facilitator, or learner. Past studies on intelligent tutoring systems [40] and teachable agents [27, 28] have examined how these roles affect behaviors and learning outcomes. Future research can investigate how various bot roles and corresponding interactions might influence group discussions, learning activities, and educational outcomes.

6 LIMITATIONS AND FUTURE WORK

Our current design has several limitations. Firstly, our current system employs a rule-based, somewhat simplistic mechanism for managing stage transitions and ensuring equal participation, while leveraging LLM exclusively for facilitating knowledge construction. The rule-based component lacks context sensitivity while the LLM-based approach faces challenges in stability and predictability. Future improvements should focus on refining the prompts and expanding the testing framework. Besides, a strategic integration of both rule-based and LLM-based methods could potentially address these shortcomings, offering a more balanced and context-aware system that enhances both the structure of conversations and the quality of knowledge construction. Secondly, this study is still preliminary, and we haven't tested whether the bot would actually affect learning behaviors, outcomes, or social dynamics. Besides, while there are many theories of collaborative learning [7, 12], we have not designed our paper based on them. In the future, we will consider conducting further studies and analysis to see how the bot affects learning and social interactions in groups and testing different theories to see how they might contribute to creating a more effective bot. Another consideration is the modality of the bot. We recognize that discussions like these often occur in-person. We will explore how this bot can be adapted for various contexts, including face-to-face interactions. Moreover, ethical design in AI is crucial, addressing issues such as trustworthiness [30], misinformation [44], and accountability and transparency [20]. To develop ethical systems, we have employed participatory and iterative design methods by involving target users—undergraduate students—in our design process to gather their ideas and voices [24]. This approach helps make AI and the design process more transparent. However, ethical design is not the primary focus of the current study. For future research, we plan to expand our design to incorporate more ethical considerations.

7 CONCLUSION

In this work, we introduced the iterative design process, the final design, and the implementation of an LLM-powered conversational AI for collaborative learning in small groups. The preliminary evaluation suggests that the overall bot performed as designed, but there are considerations regarding the timing of the bot's interventions and its roles in discussions. The evaluation also indicates that participants perceive the bot's tone and behaviors as important in their learning and engagement, although we noticed potential issues with over-reliance on the bot for autonomy in learning. We provide design implications for the integration of chatbots in collaborative learning, focusing on bot personalization in different contexts and enhancing students' awareness and autonomy in using the bot. We also suggest directions for future research, including developing an LLM-based bot using a combination of rule-based and model-based approaches and testing different collaborative learning theories with our platform.

ACKNOWLEDGMENTS

We would like to thank the undergraduate students at UCI, Tariq Brown, Neeraja Kulkarni, Kevin Zhang, Kai Campbell, and Tayven Taylor, who helped develop the system, as well as three anonymous study participants.

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A DISCUSSION TASKS

The problem was used in Wizard-of-oz study only:

Task	Write a Python function called <code>merge_sorted_lists</code> that takes in two sorted lists and returns a single merged sorted list.
Input	Two lists, <code>list1</code> and <code>list2</code> , where both lists contain integers and are sorted in non-decreasing order. The length of both lists can be from 0 to 10^5 .
Output	A single merged sorted list containing all the elements from <code>list1</code> and <code>list2</code> .

The open-ended discussion question about generative AI ethics was used in both Wizard-of-oz study and the Evaluation study:

Task	With the recent advancements in generative AI technologies, the barriers to developing AI models have significantly lowered, for example fostering a broad spectrum of applications illustrating the potential of AI to benefit society. However, the ease of access heightens the importance of ethical considerations and technical expertise among AI developers. Discuss what knowledge AI developers need to have to create ethical and effective AI applications in this new environment.
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B PROMPTS

The prompts we used are lengthy; here we present excerpts to showcase the design.

Chime-in Layer	Your name is {{botname}}. Your job is to listen to a conversation between students discussing a topic and determine when is the best time to chime in. The chat messages will be sent with the role "user" and will include the student's name and message. These students will be having a conversation about a certain topic, and they will be sharing their opinions and asking questions amongst themselves. When you notice that the conversation is going smoothly and the students have much to discuss, do not chime in. Respond only with: "...". If you notice that the conversation is repetitive, not productive, or worth discussing further, respond only with: "CHIME." If a student asks you a question using your name, respond only with: "CHIME."	Instruction
	Here is an example of a student conversation about The Civil War, where your response is illustrated as "[Assistant]: ..." or "[Assistant]: CHIME.": [Student1]: The Civil War was caused by the abolition of slavery [Assistant]: ... [Student2]: Wasn't it because southern states wanted to preserve their rights?[Assistant]: ... [Student3]: That's true but it was to preserve their right to choose to have slaves.[Assistant]: ... [Student2]: Is there anything else that could have caused the civil war?[Assistant]: ... [Student1]: um i don't know [Assistant]: CHIME.	Few-shot example
Behavior Layer	You are an AI Student Chat Facilitator. The first message you receive will be a Topic. This is the only topic of the discussion, and no other message should be used as a topic. This topic will then be used to guide a student's discussion with you as its facilitator. You are to first ask the students in the chat a question based on the prompt, and then respond to their assertions in an educative manner. You will leave enough open-endedness for the students to learn on their own. Asking follow-up questions is also a good practice. You will receive inputs marked as "user" with the Student's names and message formatted like this: "[STUDENT]: %MESSAGE HERE%" where "STUDENT" is replaced with the respective student's name. Do not try to imitate the student's responses. This means that your responses should NOT include brackets around a student's name like this: "[Student]". These are the names of the students in this discussion: {{users}} This is the topic of discussion:{{topic}}	Syntax
	Avoid repeating the same thing over and over. This is a very important part. Keep your messages relatively short.If you have multiple responses, pick the one response with the most relevance to the main question and generate a response based on that. Try to discourage comments that are not related to the topic, specifically side banter.Try to ask follow-up questions to the messages that worths discussing further.Try to build connections between students' messages and ask questions based on them.Try to challenge or critically respond to the students messages.Do not ask a question with every single response. Ask questions to invoke thought and test understanding but make sure to allow students to facilitate chats by themselves as well.	Instruction
	I can't wait for the weekend to go hiking! {}Assistant{}: That's a great weekend plan, but let's keep our focus on today's topic about historical events. {}Student{}: I think sustainable development is important in environmental conservation. {}Assistant{}: That's a great point. Can you expand on that? {}Student1{}: The economic impact of this historical event was huge. {}Student2{}: And it had major political consequences too. {}Assistant{}: Both of you make interesting points about economic and political impacts. How do you think these two aspects were interconnected during this event? {}Student{}: I believe AI should always be programmed to follow strict ethical guidelines without exception. {}Assistant{}: Ethical guidelines can sometimes be complex and situational. Does anyone have different opinions?	Few-shot Example